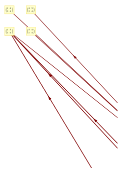
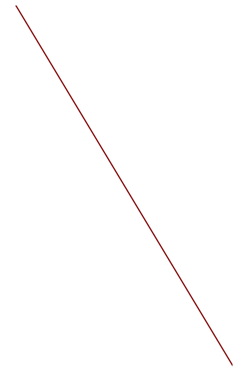
In[878]:=
```
{p, pp, q, r, sigma, tau} = {2, 1, 0, 3, {2, 3, 5, 1, 4}, {2, 1, 3, 4}};

dA = p + q + r;
dB = pp + q + r;
M1 = SCMatrix[p, pp, q, r, sigma, tau][[1]];
M2 = SCMatrix[p, pp, q, r, sigma, tau][[2]];
MT = Transpose[M1];
dz = Table[0, {Length[M1]}];
b = Table[{1, 1}, {Length[MT]}];
x = LinearProgramming[dz, MT, b, -15, Integers];
y = MT.x;
Flag = True;
For[i = 1, i ≤ Length[MT], i++,
  If[y[[i]] ≤ 0, Flag = False; Break[]];
 ];
If[Flag == False, Print["Wrong Solution"]];
x[[0]] = 0;
ptv = {};
ptvs = {};
ptg = {};
rx = 1; ry = 1;
For[i = 1, i ≤ Length[M2], i++,
  pairs = M2[[i]];
  p1 = PairPosition[M2[[i, 1]]];
  q1 = PairPosition[M2[[i, 3]]];
  v1 = {x[[p1]] / rx, x[[If[q1 == 0, q1, q1 + dA (dA - 1) / 2]]] / ry};
  p2 = PairPosition[M2[[i, 2]]];
  q2 = PairPosition[M2[[i, 4]]];
  v2 = {x[[p2]] / rx, x[[If[q2 == 0, q2, q2 + dA (dA - 1) / 2]]] / ry};
  ptg = Append[ptg, {pairs[[2]], pairs[[4]]} → {pairs[[1]], pairs[[3]]}];
  ptv = Append[ptv, {pairs[[1]], pairs[[3]]} → v1];
  ptv = Append[ptv, {pairs[[2]], pairs[[4]]} → v2];
  ptvs = Append[ptvs, {pairs[[1]], pairs[[3]]} → {p1 / rx, q1 / ry}];
  ptvs = Append[ptvs, {pairs[[2]], pairs[[4]]} → {p2 / rx, q2 / ry}];
 ];
GraphPlot[ptg, VertexCoordinateRules → ptv, VertexLabeling → True, DirectedEdges → True]
GraphPlot[ptg, VertexLabeling → True, DirectedEdges → True]
GraphPlot[ptg, VertexCoordinateRules → ptvs, VertexLabeling → True, DirectedEdges → True]
```

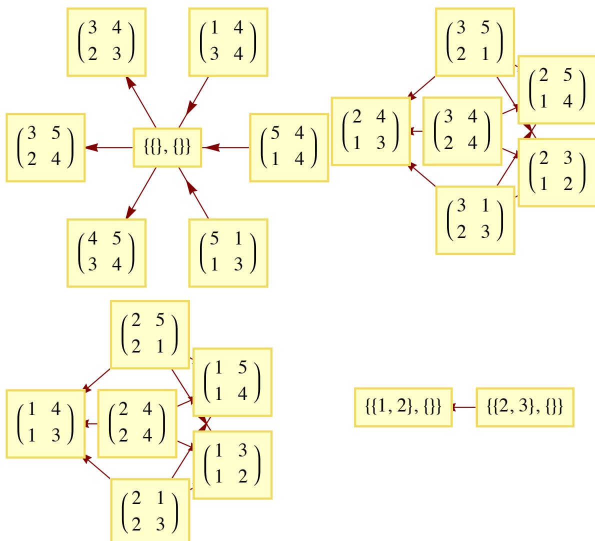
LinearProgramming::lpip :
Warning: integer linear programming will use a machine precision approximation of the inputs. ≫



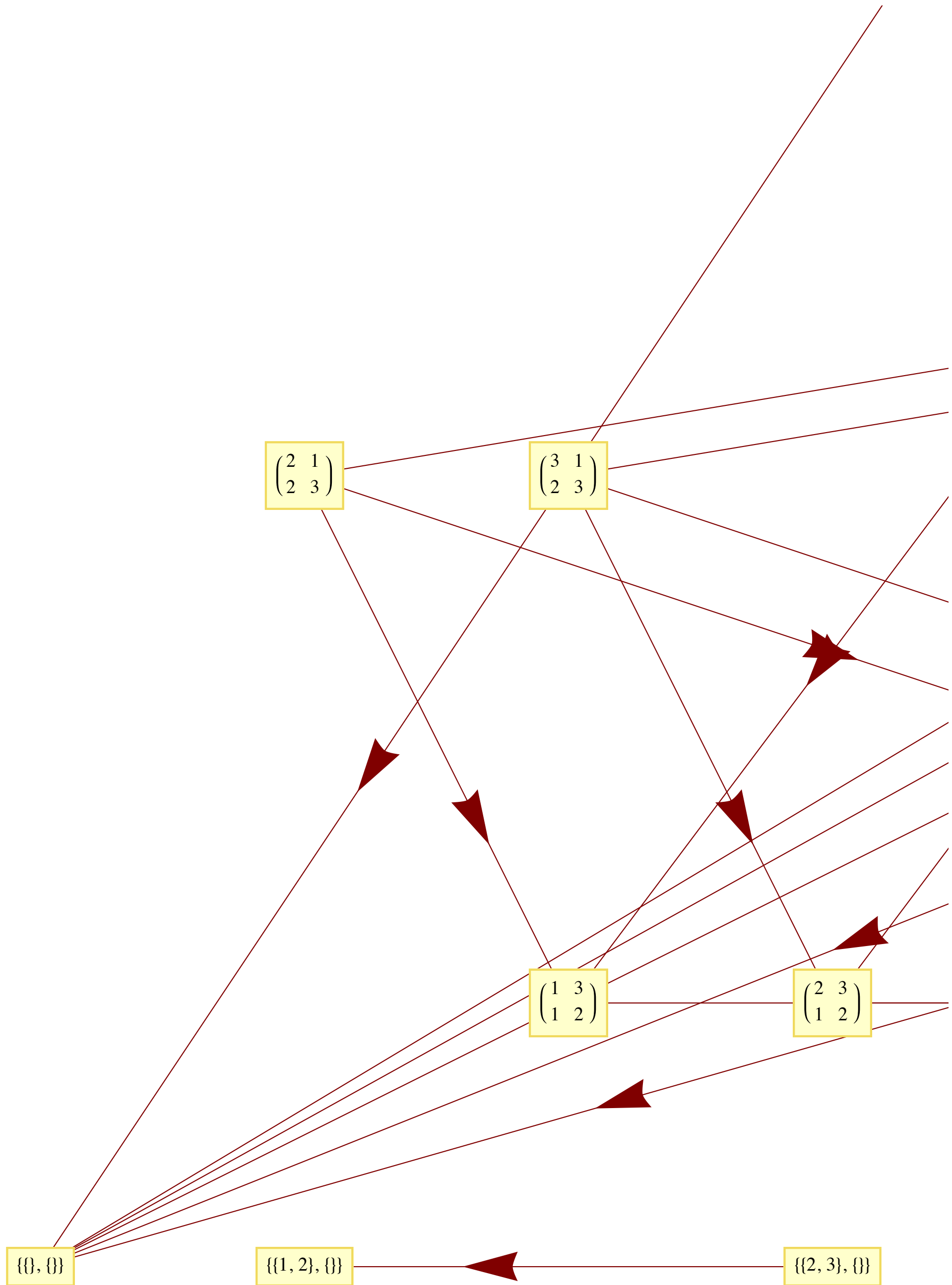
Out[697]=



Out[898]=



Out[899]=


$\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$
$\begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}$
$\begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}$
$\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$
{{}, {}}
{{1, 2}, {}}
{{2, 3}, {}}